

Project Portfolio

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SELECTED RESEARCH PROJECTS

Adaptive Multilevel Monte Carlo for Stochastic PDEs

2021–Present

Ph.D. dissertation project, University of Somewhere

[Numerical Analysis] [UQ] [C++/MPI] [SIAM J. Sci. Comput.]

Problem. UQ for subsurface flow requires statistics of solutions to elliptic PDEs $-\nabla \cdot (a(\mathbf{x}, \omega) \nabla u) = f$ with log-normal random coefficient a . Standard Monte Carlo costs $\mathcal{O}(\varepsilon^{-2} h^{-d})$ for RMSE ε .

Contribution. I designed an *adaptive* multilevel estimator that selects mesh levels $\{h_\ell\}$ and sample counts $\{N_\ell\}$ online by minimising cost subject to a variance budget, using a Gaussian process surrogate to predict level contributions.

- Proved optimal complexity $\mathcal{O}(\varepsilon^{-2} |\log \varepsilon|^2)$ without *a priori* knowledge of coefficient regularity.
- Implemented in C++17/MPI with FEniCS and PETSc; near-linear weak scaling to 512 MPI tasks on Stampede2.
- $3\times$ wall-clock speedup over standard MLMC on a 3-D Darcy flow benchmark with Matérn-3/2 coefficient.

Key result: Paper accepted, *SIAM J. Sci. Comput.* 2024. Code: github.com/yourhandle/adaptive-mlmc.

Sharp Convergence Analysis of Nonconvex SGD

Summer 2022

Research internship, National Lab

[Optimization Theory] [ML Theory] [Python/JAX] [arXiv preprint]

Problem. Convergence guarantees for SGD on nonconvex objectives remain loose; prior bounds require bounded gradients or structural assumptions rarely satisfied in practice.

Contribution. I established a sharp $\mathcal{O}(k^{-1/2})$ rate for $\mathbb{E}[\|\nabla f(x_k)\|^2]$ under (L, σ) -smoothness — strictly weaker than standard L -smoothness — via a Lyapunov function accounting for batch-size-dependent variance.

- Derived tight lower bounds confirming $\mathcal{O}(k^{-1/2})$ is unimprovable without additional curvature information.
- Validated on CIFAR-10/100; empirical convergence matched predicted rates across 6 architectures.
- Compared AdaGrad, Adam, SGD-M; identified regimes where adaptive methods provably dominate.

Key result: Preprint arXiv:2401.00000; under revision for NeurIPS 2024.

Sparse-Grid Collocation for High-Dimensional Bayesian Inversion

Spring 2023

Independent study / course project

[Bayesian Inference] [Sparse Grids] [Python] [SciPy/Chaospy]

Problem. Bayesian inversion of distributed parameter fields leads to posteriors on high-dimensional ($d = 50$ – 200) spaces; MCMC is prohibitive when each forward solve costs ~ 1 CPU-minute.

Approach. I built a surrogate-accelerated pipeline: a Smolyak sparse-grid interpolant approximates the log-likelihood offline ($\sim 10^3$ PDE solves), then Metropolis-Hastings samples the surrogate at negligible cost.

- $< 5\%$ relative error in posterior mean and variance for a 50-D groundwater conductivity inversion.
- Clenshaw–Curtis nodes reduce interpolation error by $2\times$ over Legendre nodes for smooth likelihoods.
- Reusable Python module with FEniCS interface; tested on two published benchmarks.

ADDITIONAL PROJECTS

Finite-Element Solver for Incompressible Navier–Stokes

2019–2020

Senior honors thesis, State University

[FEM] [Fluid Dynamics] [C++] [deal.II]

Taylor–Hood ($\mathbf{P}_2/P_1$) mixed FEM for 2-D incompressible flow using deal.II; second-order velocity convergence verified via manufactured solutions. Lid-driven cavity flow at $\text{Re} = 400, 1000, 3200$ matches Ghia et al. (1982) to 4 significant figures.

Computer-Assisted Hopf Bifurcation Proof

Summer 2019

REU project, Another University

[Dynamical Systems] [Interval Arithmetic] [MATLAB/Intlab]

Used INTLAB to rigorously enclose Jacobian eigenvalues of a 3-species Lotka–Volterra system and certify a sign change in the real part, constituting a computer-assisted proof of Hopf bifurcation at $\mu^* = 0.7134 \pm 10^{-4}$.

TECHNICAL STACK

Domain	Tools & Methods
Numerical PDE	FEniCS, deal.II, PETSc, Firedrake, FDM, FEM, Spectral methods
HPC	C++17, MPI, OpenMP, SLURM, vectorised NumPy, JAX JIT
UQ / Stoch.	MLMC, Sparse Grids, Polynomial Chaos, MCMC, Kalman filters
Optimisation	SciPy, JAX (autodiff), L-BFGS, proximal methods, ADMM
ML	PyTorch, scikit-learn, Gaussian processes (GPyTorch)
Languages	Python, C/C++, MATLAB, Julia, R, Bash
